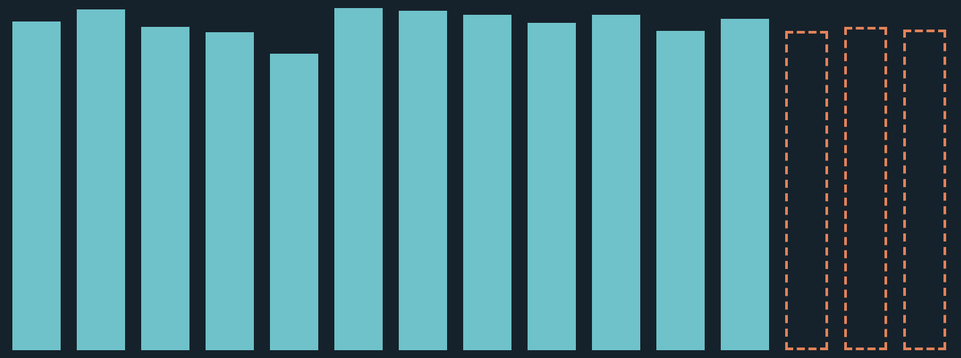
CAPSTONE DEFENSE

Kenya Retail Sales & Customer Intelligence

Inflation-aware demand forecasting



Monthly units sold, Oct 2025 to Sep 2026 (solid), and the three forecast months (dashed)

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LuxDevHQ Data Science & Analytics Capstone

WHY THIS PROJECT

What should a store manager stock next?

Five research questions, answered with Python, SQL and Power BI on twelve months of retail data.

1. How do sales and demand vary over time, by category and by weekday?
2. Which products show the largest changes in demand?
3. Do promotions or inflation show a detectable link to demand?
4. Can a simple model forecast monthly units for the next three months?
5. What can and cannot be concluded about inflation sensitivity?

THE DATA

Synthetic sales plus official KNBS CPI

Table	Rows	What it holds
sales	15,000 lines	5,617 transactions
customers	1,000	4 segments, 5 age groups
products	100	4 categories, KSh price and cost
stores	15	5 counties, 3 stores each
promotions	15	8% to 25% discounts
public holidays	25	national holidays
national CPI (KNBS)	141	monthly index since 2015

Only the CPI is official data.

Results from the synthetic transactions demonstrate an analytical workflow. They are not evidence of real Kenyan retailer or consumer behaviour.

HEADLINE RESULTS

The year at a glance

17.06M

Net sales in KSh, after discounts

36,151

Units sold across 5,617 transactions

23.16%

Estimated gross profit margin (KSh 3.95M profit)

994

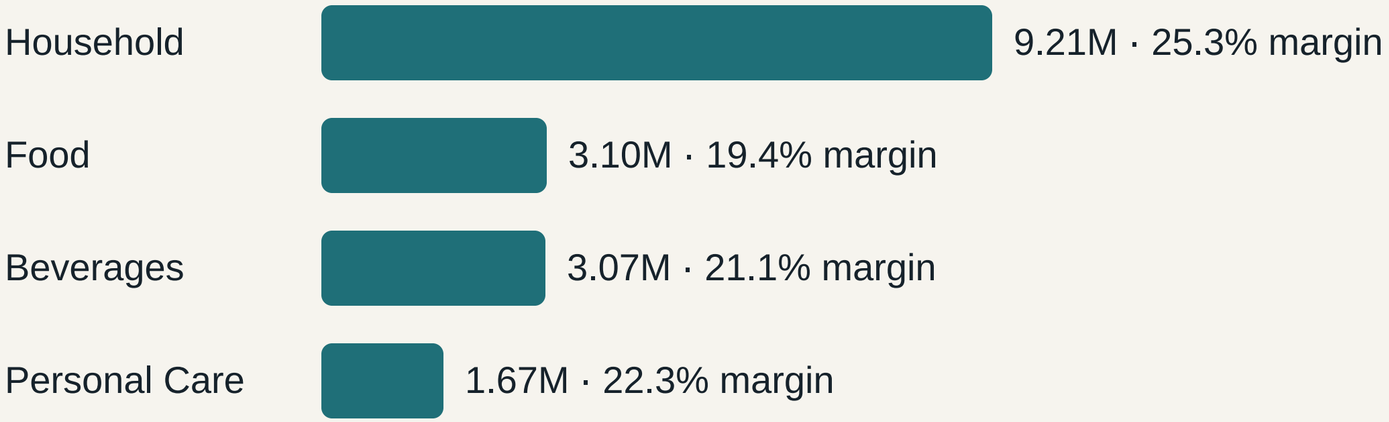
Active customers out of 1,000

Profit is estimated from catalogue unit costs. The transactions are synthetic.

WHAT SELLS

Household earns 54% of sales from 9,129 units

Net sales in KSh millions and estimated margin, by category



Margin = (net sales − unit cost × quantity) ÷ net sales.

58%

of net sales come from the top 10 products

14,227

units of Food: the most units, but the lowest margin

KSh 3.53M

from one product, Microwave Oven 20L (313 units)

WHO BUYS

Customer segments behave almost identically

Net sales per customer in KSh, by segment (bars start at zero)



The labels New, Occasional, Lapsed and Regular do not predict spend or visit frequency in this dataset.

5.6 to 5.7

transactions per customer, in every segment

64 days

average since last purchase for “Lapsed” customers, against 62 for “Regular”

0.92M–1.43M

KSh net sales across the 15 stores; the top store holds only 8%

STATISTICAL TESTS

Most tests found no detectable effect

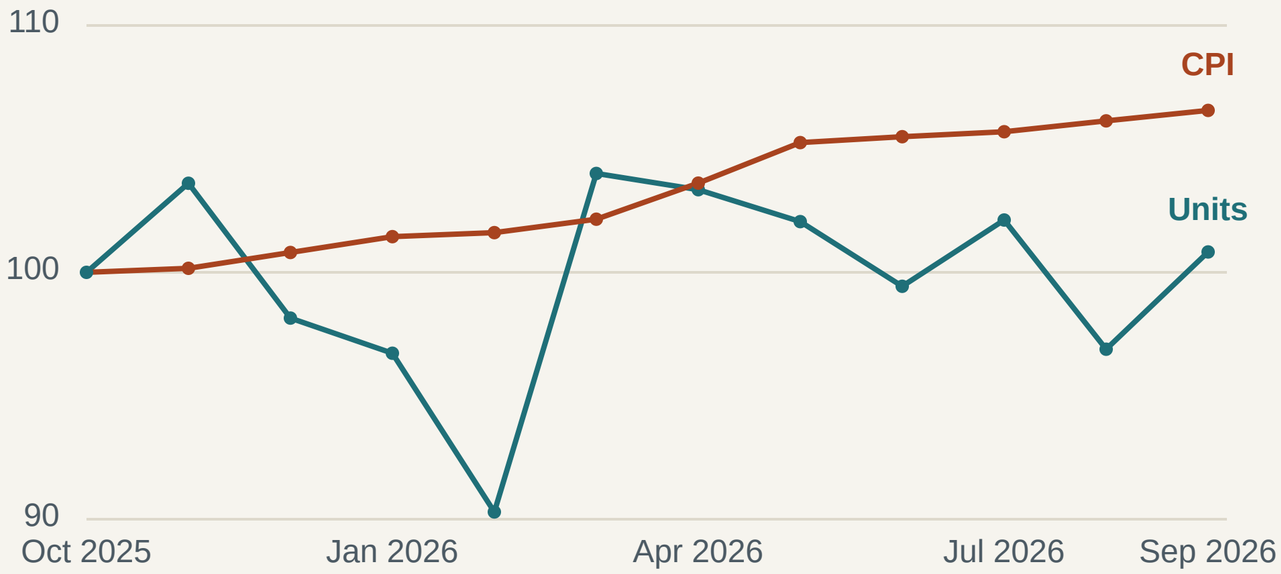
Question	Test	Result
Categories differ in sales value?	Kruskal–Wallis	Yes: H = 846.35, p < 0.001
Weekdays differ in daily sales?	Kruskal–Wallis	No clear evidence: p = 0.405
Promotion days vs other days?	Mann–Whitney	No clear evidence: p = 0.085; 95% CI −8.5k to +2.7k KSh/day
Monthly units vs CPI	Spearman	ρ = 0.028, p = 0.931
Monthly sales value vs CPI	Spearman	ρ = 0.329, p = 0.297
Personal Care units vs CPI	Spearman	ρ = 0.588, p = 0.044 (nominal)
Product demand trends	Spearman	7 before correction, 0 after Benjamini–Hochberg

With only 12 monthly points, a missing effect is not proof that no effect exists.

INFLATION

CPI climbed 6.6% while demand stayed flat

Monthly units sold and CPI, both indexed to October 2025 = 100



+6.6%

CPI rise, from 146.84 in Oct 2025 to 156.47 in Sep 2026

ρ = 0.03

Spearman correlation of monthly CPI with units (p = 0.93)

No causal claim

Twelve months and one national index can show co-movement at most, not cause.

FORECASTING

Three models, one chronological test

Model	MAE	RMSE	WAPE
Random Forest	77.74	82.40	2.58%
Naive (last month)	119.33	123.40	3.95%
Linear Regression	146.81	172.64	4.87%

Lower is better. WAPE is total absolute error divided by total actual units over July to September 2026.

Inputs: units from the previous three months plus the month number. The Random Forest depth is capped at 2 to limit overfitting. Honest note: always predicting the training mean scores 2.25%, so the forest is effectively the mean.

TRAIN

Jan to Jun 2026, six months

TEST

Jul to Sep 2026, three months, predicted one month ahead at a time

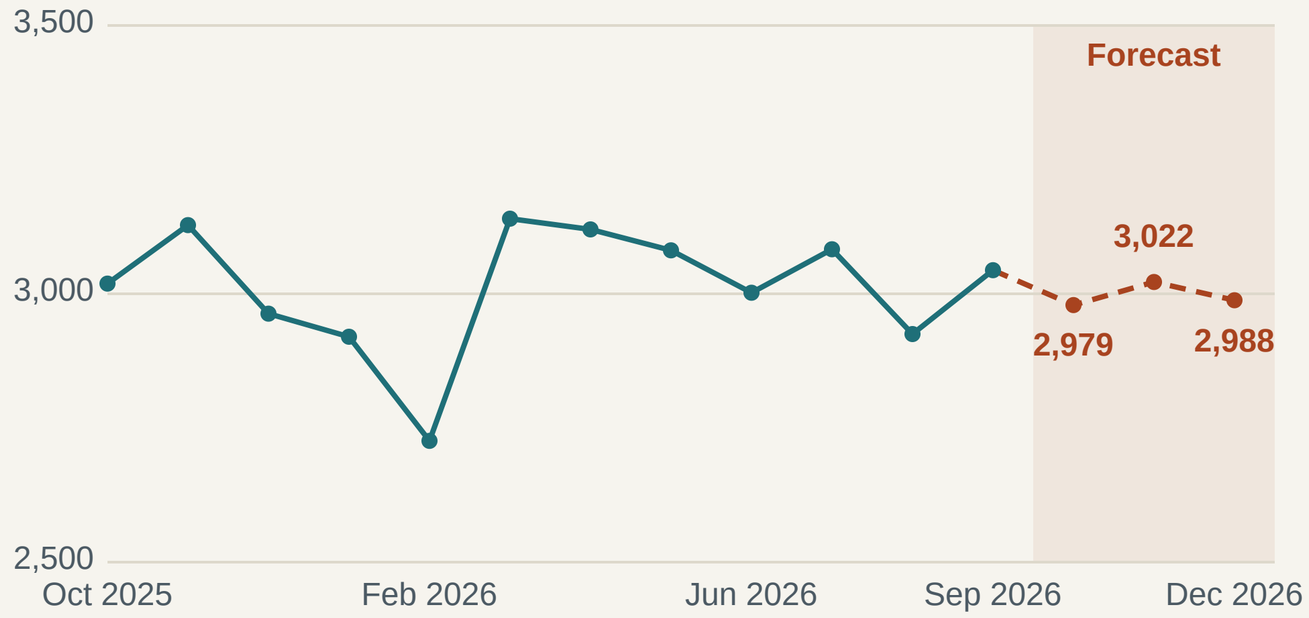
FORECAST

Oct to Dec 2026, Random Forest, each month feeding the next

THE FORECAST

Next quarter: about 3,000 units a month

Monthly units sold: actual (solid) and Random Forest forecast (dashed)



8,989

units forecast for Oct to Dec 2026

2.58%

WAPE on the three test months

Flat by design

Twelve months cannot teach seasonality. Use it as a planning baseline, with a 3% to 5% buffer.

THE DASHBOARD

A four-page Power BI dashboard for the store manager

PAGE 1

Executive Overview

How much are we selling, and what comes next?

KPI cards, monthly net sales, actual units against the forecast

PAGE 2

Product & Store Performance

What sells, and where?

Top and bottom 10 products, category and store sales, filters

PAGE 3

Customer Intelligence

Who buys, and how often?

Segments, age groups, 97.9% repeat rate, 736 customers active in the last 90 days

PAGE 4

Demand Forecast & Inflation

What should we plan for?

Three-month forecast, model test, CPI beside demand, limitations

Built from the cleaned CSVs and the forecast exports. Forecast months sit in separate tables and are never counted as actual sales.

LIMITATIONS

What this project cannot claim

Synthetic transactions

A demonstration of the workflow, not evidence of real shopper behaviour. Profit is estimated.

Only 12 months

Trained on 6 months and tested on 3. Each CPI test uses 12 points.

A flat forecast

No seasonality or promotion effect can be learned from a single year.

Constant prices

Each product has one catalogue price, so price sensitivity cannot be measured.

Promotions by date only

PromotionID was dropped from sales, so lift cannot be tied to a campaign.

Source data quirks

1,911 lines predate the customer’s signup date, and the age label “44-54” is probably “45-54”.

RECOMMENDATIONS

What the store manager can do next

Each action comes from a finding in this project.

1. Plan stock at about 3,000 units a month, with a 3% to 5% buffer
2. Protect availability of the top household appliances: 10 products drive 58% of sales
3. Review very low-value items such as Avocado and Salt for shelf space or bundling
4. Investigate the two lowest stores, Coast Retail 03 and Metro Retail 02
5. Rebuild segments from recency and frequency, and collect real prices and promotion IDs